

JEE MAIN REVISION MASTER: HYDROGEN AND IT'S COMPOUNDS 2026 0 QUESTIONS 0% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet Hydrogen And Its Compounds - Core Application	9 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Hydrogen And Its Compounds - Core Application (High Yield — 9 PYQs)

Core Principle: Key problem pattern in Kcet Hydrogen And Its Compounds.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Which one of the following properties is generally not applicable to ionic hydrides?..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "The correct decreasing order of boiling point of hydrogen halides is..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "The composition of water gas is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#2)

The correct decreasing order of boiling point of hydrogen halides is

- (A) $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$
- (B) $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$
- (C) $\text{HF} > \text{HI} > \text{HBr} > \text{HCl}$
- (D) $\text{HI} > \text{HF} > \text{HBr} > \text{HCl}$

Correct Answer: None

Solution Analysis: HF is hydrogen bonded, thus it has highest boiling point. The remaining hydrogen halides are gaseous and their boiling points depend on the van der Waals' forces. Greater the molecule mass, greater will be the van der Waals' forces. Hence higher is the boiling point. Thus, the correct order is $\text{HF} > \text{HI} > \text{HBr} > \text{HCl}$.